

Domino Neuron Relay

Make a nerve signal survive a gap.

Win condition: Fastest reliable signal with best explanation.

THE SCIENCE YOU ARE USING

Signals travel in a chain. A line of falling dominoes is a model of a nerve signal: each one knocks the next, so the signal travels without any single domino moving far. A real neuron passes a signal down its length the same way, one step triggering the next.

Gaps and insulation. If the gap between dominoes is too large, the signal dies. Neurons face the same problem at synapses. A myelin sheath speeds signals along; you model gaps and insulation to keep the relay fast and reliable.

HOW TO BUILD AND RUN IT

- 1 Build a baseline track.** Set dominoes at an even spacing and time one full run.
- 2 Introduce a gap.** Open a larger gap and see whether the signal still crosses.
- 3 Bridge the gap.** Add a bridge piece, like a synapse helper, so the signal survives.
- 4 Speed it up.** Adjust spacing to make the relay faster without breaking the chain.
- 5 Explain the analogy.** Connect your design choices to neurons, synapses, and myelin.

Safety: Low risk. Allergy: None.

MATERIALS

Dominoes	per team
Rulers	per team
Tape	shared
Index cards	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Transmission success	35
Gap challenge	25
Explanation	20
Redesign	20
Total	100